Journal of The
Electrochemical Society

The Electrochemical Society
Advancing solid state & electrochemical science & technology

Induction-Diffusion-Adsorption Process of Malachite Green for Cobalt Superconformal Electrodeposition

To cite this article: Xiaochuan Ma *et al* 2023 *J. Electrochem. Soc.* **170** 022503

View the article online for updates and enhancements.

You may also like

- Rutile phase dominant TiO2 formed by thermal treatment and its high photocatalytic activity under narrow spectrum ultraviolet light emitting diodes
Kalithasan Natarajan, Thillai Sivakumar Natarajan, Hari C Bajaj et al.
- A green approach for the preparation of ZnO@C nanocomposite using agave americana plant extract with enhanced photodegradation
Mansab Ali Jakhrani, Aneela Tahira, Muhammad Ali Bhatti et al.
- Construction of silver-doped graphitic carbon nitride integrated zeolitic imidazolate framework as an ultra-sensitive probe for colorimetric detection application
Rustem R Zairov, Abdallah M Elgorban, Rafid Kamal Jameel et al.

ECC-Opto-10 Optical Battery Test Cell:
Visualize the Processes Inside Your Battery!

EL-CELL®
electrochemical test equipment

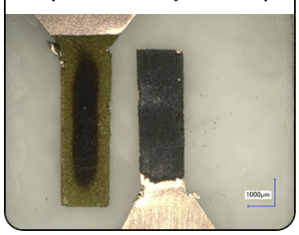
- **Battery Test Cell for Optical Characterization**
  Designed for light microscopy, Raman spectroscopy and XRD.
- **Optimized, Low Profile Cell Design (Device Height 21.5 mm)**
  Low cell height for high compatibility, fits on standard samples stages.
- **High Cycling Stability and Easy Handling**
  Dedicated sample holders for different electrode arrangements included!
- **Cell Lids with Different Openings and Window Materials Available**



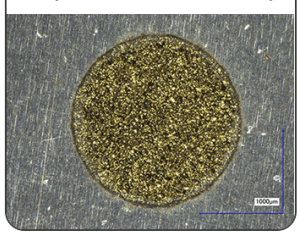
Scan me!

Contact us:
+49 40 79012-734
sales@el-cell.com
www.el-cell.com

Sample Test (Side-by-Side Setup)



Sample Test (Face-to-Face Setup)



This content was downloaded from IP address 219.217.246.195 on 02/10/2025 at 04:46

Journal of The Electrochemical Society, 2023 **170** 022503
1945-7111/2023/170(2)/022503/7/$40.00 © 2023 The Electrochemical Society ("ECS"). Published on behalf of ECS by IOP Publishing Limited

Induction-Diffusion-Adsorption Process of Malachite Green for Cobalt Superconformal Electrodeposition

Xiaochuan Ma,[1] Yaqiang Li,[1] Penghui Ren,[1] Peixia Yang,[1] Jinqiu Zhang,[1] Maozhong An,[1,z] and Wenquan Lv[2,z]

[1]*State Key Laboratory of Urban Water Resource and Environment, School of Chemistry and Chemical Engineering, Harbin Institute of Technology, Harbin 150001, People's Republic of China*
[2]*State Key Laboratory of Solid Lubrication, Lanzhou Institute of Chemical Physics, Chinese Academy of Sciences, Lanzhou 730000, People's Republic of China*

Cyclic voltammetry (CV), chronopotentiometry and quartz crystal microbalance were used to study the effect of Malachite Green (MG) on bottom-up growth of cobalt. The addition of MG not only inhibited the reduction of Co^{2+} but also accelerated the hydrogen evolution process. The metal reduction process was enhanced as well as the current efficiency was higher with the increase of pH. The bottom-up growth of cobalt can be achieved with the potential of −1.20 V. Induction-diffusion-adsorption superconformal electrodeposition mechanism was proposed to illustrate the synergistic effect of MG and pH on the growth of cobalt. On one hand, the MG concentration gradient and pH gradient could be formed to achieve a growth rate difference within blind vias so as to obtain bottom-up growth (diffusion-adsorption effect). On the other hand, the growth rate difference was intensified due to the accelerating effect of MG on HER so as to boost the cobalt superfilling (induction effect).
© 2023 The Electrochemical Society ("ECS"). Published on behalf of ECS by IOP Publishing Limited. [DOI: 10.1149/1945-7111/acb9bf]

Manuscript submitted November 24, 2022; revised manuscript received January 29, 2023. Published February 15, 2023.

Supplementary material for this article is available online

Since the copper plating process, namely Damascus Copper Plating process, has been proposed by IBM in order to achieve metal interconnect integrated circuits (IC), it has attracted a great deal of attention in terms of electroplating technology in electronic devices with metal deposition of copper interconnect line.[1] Up to now, the damascene plating technology is still leading technology for the production of large scale integration (LSI) devices.

However, within the manufacturing process of back-end-of-line (BEOL), the width of interconnect lines has been downscaled to nano level with the density of LSI devices becoming more and more compressed according to the Moore's Law. Resulting in the increase of the copper resistivity and RC delay, which has become the key factors for restricting the performance of the integration devices.[2,3] For example, the resistivity of a 10 nm-wide Cu interconnect line is more than ten times higher than that of bulk copper due to the severe electrons scattering and grain boundaries scattering.[4,5]

As a result, it is of great importance to screen metals with low electron mean free path in the replacement of Cu in order to improve the property of IC. Among these metals with relatively low mean free path, Co was investigated widely as interconnect line after balancing the production cost and IC property.[6–11] Since cobalt not only is of low fine-line resistivity, but also is of strong electro-migration resistance.[12,13]

Applying electrodeposition method to prepare Co thin films is a cost-friendly, fast, and easy-operating method, and has been applied in many fields.[14–17] Achieving bottom-up growth of cobalt is the key to manufacturing the interconnect line, while the superconformal electrodeposition of cobalt can be obtained by exploring additive,[18] and optimizing the electrodeposition parameters, including the composition of bath solution,[19] current density,[20] pH[21] and so on. Besides, Co film with different structure and crystal phase can be obtained by changing electrodeposition parameters.[22] Generally, the Watts-based electrolyte or buffered sulfate electrolyte can be used to deposit Co film, in which boric acid has been added to stabilize the increasing pH in plating bath caused by polyborates and primarily triborate and occupy block surface sites to inhibit HER as well as facilitate cobalt deposition by the surface complex with cobalt.[23–25] In order to achieve bottom-up growth of cobalt deposit, appropriate additives should be introduced into plating bath to achieve growth rate difference in the blind vias.[26–28] And, this difference is consistent with the diffusion-adsorption mechanism.[29–32]

Josell et al.[28] realized the superconformation filling of TSVs with high aspect ratio by adding polyethyleneimine (PEI) as suppressor into the plating solution, which is composed of cobalt sulfate, cobalt chloride and boric acid. The deposition mechanism is due to the occurrence of S-NDR characteristics after the breakdown of the adsorption of PEI on the cathode surface, in which active-passive transitions are formed in TSVs with the applied voltage. Hydrogen induction deactivation mechanism was proposed to explain the influence of additive XF on superfilling of cobalt by Wu et al.[33] The growth rate different within blind vias was ascribed to the fact that the deactivation rate of additive is higher than the compensatory rate by mass transport. Huang et al.[34] added the dioxime additive as suppressor to achieve Co defect-free filling. $Co(DMG)_2$ complex can be formed between dioxime and cobalt ions so as to inhibit the Co reduction. When the voltage rises to the breakdown threshold, Co begins to deposit on the cathode surface. Under the influence of convection, an obvious inhibition gradient was formed during the deposition process, which is consistent with S-NDR mechanism.

In this study, Malachite Green (MG), was selected as a suppressor for cobalt electrodeposition. It has been verified that MG adsorbed on the cathode surface to inhibit the reduction of copper and titanium.[35] And the influence of MG on Co metal deposition process have also been discussed before,[36] so this paper mainly studies the influence of MG adsorption on the hydrogen evolution process. The electrochemical tests, including cyclic voltammetry (CV) and linear sweeping voltammetry (LSV), were applied to research the inhibition ability of MG on the deposition process of cobalt, while the influence of MG on HER was also studied by EQCM tests. Superconformal electrodeposition of cobalt was achieved and induction-diffusion-adsorption mechanism was proposed to illustrate the synergistic effect of MG and pH.

Experimental

Materials and solutions.—The cobalt solution consisted of 0.4 mol l^{-1} $CoSO_4{\cdot}7H_2O$, 0.5 mol l^{-1} H_3BO_3, 10 mM Cl^- and 20ppm 3-Mercapto-1-propanesulfonic acid (MPS) at 298 K. The pH of the solution was adjusted to 4.0 with diluted H_2SO_4 and NaOH solution. All solutions were prepared using analytical grade reagents with ultrapure water. Concentrated Malachite Green (MG) solution (1,000 ppm) was prepared from the analytical pure powder of MG.

[z]E-mail: mzan@hit.edu.cn; wqlv@licp.cas.cn

The base electrolyte consists 0.4 mol l^{-1} $CoSO_4 \cdot 7H_2O$, 0.5 mol l^{-1} H_3BO_3, 10 mM Cl^- and 20 ppm MPS for electrochemical analysis. The virgin makeup solution (VMS) for pre-plating Co was composed of 0.4 mol l^{-1} $CoSO_4$, 0.5 mol l^{-1} H_3BO_3.

Electrochemical test.—The electrochemical behavior was investigated by cyclic voltammetry (CV) and linear sweeping voltammetry (LSV) from a three-electrode cell by Gamry (Interface 1010E) potentiostat with the scan rate of 5 mV s^{-1}. A Mercury sulfate electrode (MSE, containing saturated K_2SO_4) was used as the reference electrode, and all tests were referred to MSE. A Co-coated Pt used as rotating disk electrode (RDE) of 0.5 cm diameter (0.196 cm^2) was used as cathode. The thickness of Co coating was around 300 nm. A Pt foil was counter electrode with a surface area much larger than the cathode. The cell was purged by high-purity nitrogen for 0.5 h before all the electrochemical test.

The Gamry (EQCM 10 M) with a 5 MHz gold sensing electrode (1.54 cm^2) was used for electrochemical quartz crystal microbalances (EQCM) experiments. Au wafer was plated from VMS for 5 min as working electrode and Pt foil as the counter electrode. The current density (j_{EQCM}, mA cm^{-2}) for the Co deposition was calculated from the equation.[37]

Electroplating.—The electroplating bath contained 0.4 mol l^{-1} $CoSO_4$, 0.5 mol l^{-1} H_3BO_3, 10 mM Cl^-, 20 ppm MPS and 10 ppm MG, and pH was adjusted to 4.0 at 298 K with the stirring rate of 500 rpm. A three-electrode cell was used for cobalt deposition process with Co-coated blind via as cathode, 99.99% Co foils as anode and MSE as the reference electrode. The depth of microvias used in this work is 100 μm, while the diameter is 50 μm, 100 μm, 150 μm. The blind via was pre-plating at −1.25 V for 5 min, after which nitrogen was used to dry blind vias. The blind vias were immersed into 10% H_2SO_4 solution, and then the vias were rinsed with large amounts of deionized water (with a resistivity of 18.2 MOhm-cm). Pre-wetting with ethanol was carried out before being immersed in the bath. The morphology of cobalt coating was obtained with a Helios (Nanolab 600i) scanning electron microscope (SEM) and an Olympus (DSX 510) optical microscope.

Results and Discussion

The effect of MG on cobalt reduction process.—The electrochemical behaviors of MG were obtained from CV in order to study the influence of MG on cobalt reduction process. The influence of MPS on electrochemical behavior can be regarded as an accelerator for the electrodeposition of cobalt with a depolarization effect according to LSV analysis in Fig. S1. Figure 1 shows the CV curves of 10ppm MG in the bath with the different rotation rate so as to study the influence of mass transport. It can be seen that the current density begins to increase sharply when potential reaches to −1.24 V (vs MSE) without rotating. When the rotation rate increased from 0 rpm to 800 rpm, the threshold potential of Co ion reduction shifted negatively from −1.24 V to −1.31 V (vs MSE) on the negative-going sweep, which can be attributed to the adsorption of the MG[38] and the change of surface coverage. The adsorption MG can be used as site-blocking by decreasing the active surface area for cobalt ions reduction. And the current density of the reduction process gradually decreases with the increasing rotation rate, due to the faster replenishment of MG from the bath to the cathode surface. In addition, the current loop at around −1.26 V was observed, demonstrated that the overpotential of cobalt nucleation is large. The slightly increase of current at about −1.15 V to −1.24 V was mainly attributed to the reduction of hydrogen ion and H_2O.[39,40] No metal deposits were found on the cathode surface (not shown here). In this case, the total current density vs potential curve exhibits S-shape negative diffierential resistance (S-NDR) behavior,[28,41,42] which attributes to the breakdown of the blocking inhibitor layer (MG layer).

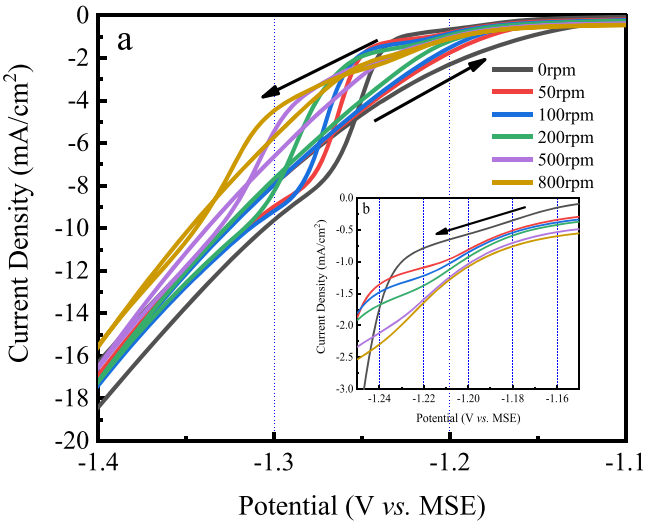


Figure 1. The CV curves of deposition solution at different rotating rate from the open circuit potential to −1.50 V: (a). the CV curves at different rate with 10 ppm MG, the detailed figures within −1.25 V to −1.15 V was shown in the inserted figure (b). The Co-coated Pt RDE used as the cathode. The iR-drop of cell resistance was compensated from electrochemical impedance spectroscopy. Scan rate was 5 mV s^{-1}.

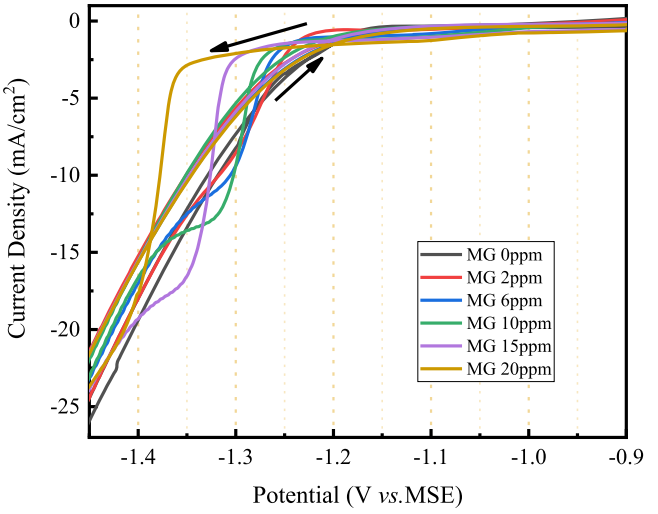


Figure 2. The CV curves of different concentrations of MG in the base electrolyte at 100 rpm. Co-coated Pt used as cathode, Co foil used as anode. Scan rate of 5 mV s^{-1}.

It should be noted that the current density for H^+ reduction also changed between −1.16 V and −1.24 V with rotation speed increased from 0 rpm to 800 rpm shown in Fig. 1(b). At −1.16 V, the current density for H^+ reduction reached to −0.3 mA cm^{-2} at 0 rpm and −0.7 mA cm^{-2} at 800 rpm, which is attributed to the acceleration of H^+ reduction with the increasing speed. Besides, the slope before reaching −1.19 V is different from that after reaching −1.19 V, illustrating that the partial current was attributed to reduction of hydrogen ion (reaction 1[43]) before reaching to −1.19 V, as well as the current caused by the water reduction (reaction 2[44]) when potential was within −1.19 V to −1.22 V.[24,43]

$$2H^+ + 2e^- \rightarrow H_2 \quad [1]$$

$$2H_2O + 2e^- \rightarrow H_2 + 2OH^- \quad [2]$$

The CV curves of different concentrations of MG on electrochemical behavior were tested, as shown in Fig. 2. It can be seen that

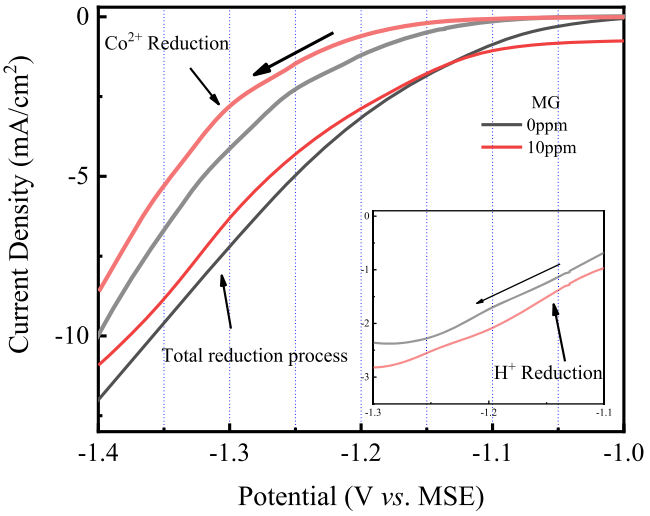


Figure 3. EQCM-derived cobalt current response at pH 4, with (red) and without (black) MG in the base electrolyte. There is no stirring and Co-coated Au as cathode, Pt wire as anode. The hydrogen evolution current was collected from subtracting the EQCM-derived current density from the total voltammetric current density. Scan rate is 5 mV s^{-1}.

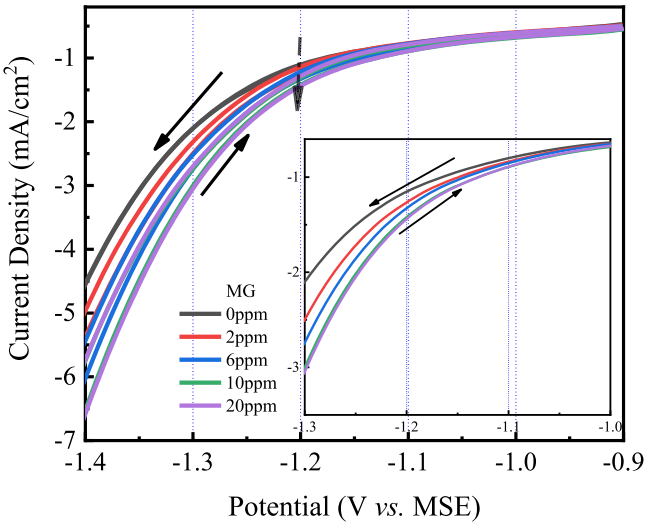


Figure 4. Cyclic voltammetry of hydrogen evolution in base electrolyte containing various concentration of MG with Co-free replaced by K_2SO_4 on the Co-plating Pt cathode at 500 rpm. Pt foil used as anode. Scanning rate is 5 mV s^{-1}.

the significant hysteretic voltammetric curves appeared after MG was introduced into the electroplating bath even though the concentration is quite low (2 ppm), demonstrating that the additive MG is of excellent inhibition ability for the reduction of cobalt. It should be noted that the strong inhibition effect might be caused by strong adsorption and coordination between MG and cobalt ions studied in our previous work.[45] Besides, the breakdown potential shifted negatively from about −1.25 V to lower than −1.40 V with the MG concentration increasing from 2 ppm to 20 ppm, while there is a relationship between the increase of threshold potential and the concentration of MG, illustrating that the threshold value is related to the adsorption and transport of MG. Meanwhile, the S-shape negative differential resistance (S-NDR) can be observed when the MG concentration is higher than 10 ppm since the current density rises sharply when the cathode potential is lower than the breakdown potential. Furthermore, the current density of H^+ reduction was also accelerated with the increase of MG.

The acceleration of MG on HER.—Because the standard reduction potential of Co^{2+} to Co metal (−0.28 V vs NHE) is more negative than that of H^+ reduction to H_2 (0 V vs NHE), the side reaction (hydrogen evolution reaction, HER) would occur inevitably and have a negative effect on the current efficiency of Cobalt deposition in an acidic bath. As a result, the influence of MG on HER should be taken into account respectively. The influence of MPS on HER has been investigated and MPS has little effect on HER, as shown in Figs. S2 and S3.

EQCM was used to separate the Co deposition current and HER current respectively, as shown in Fig. 3. It can be seen that under the condition of none convection, the total current density began to increase when the potential scanning to −1.10 V after MG was added into the plating bath, lower than that obtained without MG. Likewise, there is a similar trend for the current density of the Co reduction at about −1.14 V, which was attributed to the absorption of suppressor MG and the CoMG complexes. The decreased current density of HER with a potential range from −1.10 V to −1.30 V, shown in the inset, was mainly due to the reduction current of H_2O or H^+. At lower overpotential, the current density of the proton reduction is higher than that of Co reduction, which means that the main electrochemical reaction is hydrogen evolution process, consistent with the thermodynamic result. In other words, the current density of HER was boosted with the existence of MG. So, it can be concluded that MG can suppress the reduction of Co^{2+} ions but boost the HER.

What's more, voltammetry curves were used to further study the influence of MG on HER without Co ion in the bath. The solution contains of the same composition of plating bath except that the $CoSO_4{\cdot}7H_2O$ was replaced by K_2SO_4 in order to match the conductivity of electrolyte. Co-seeded platinum electrode was selected as the cathode was shown in the Fig. 4. It can be seen that the current density increases with the increase of additive concentration at about −1.0 V. The process of hydrogen evolution was significantly accelerated even though 2 ppm MG added into plating bath. The accelerating effect on HER can be enhanced with the increase of MG concentration. The MG could absorb on the cathode surface as active sites to promote the HER, which is consistent with EQCM analysis (Fig. 3). It should be noted that the current density obtained in EQCM analysis is slightly different from that of cyclic voltammetry analysis, which is due to the fact that rotation electrode was applied in cyclic voltammetry tests while there was none convection in EQCM test.

The process of hydrogen evolution was significantly accelerated after MG was added into plating bath[46] and the hydrogen evolution current increased with the increasing concentration of MG. In other words, the higher coverage of adsorption MG on electrode surface, the higher current density for proton reduction (Fig. 1). Furthermore, the hydrogen evolution current obtained is lower than that collected from EQCM, which indicated the complex CoMG might also act as a catalyst for HER.[47,48] In summary, the MG absorbed on the cathode surface to inhibit cobalt ions by occupying the active site on the surface but to boost the HER during the cathodic reduction.

According to the above description, the addition of MG inhibited the deposition of cobalt, but accelerated HER process at the same time at lower overpotential. This indicates that the MG will strongly adsorb on the cathode surface and occupy the active site, which plays an inhibitory role on cobalt deposition. However, MG occupying the active site is beneficial to the reduction of H^+/H_2O as a catalyst.

The acceleration of MG on HER will promote the appearance of hydroxide at the electrode surface, the intermediate formation of $CoOH^+$ or $Co(OH)_2$ might also been appeared during the deposition of cobalt metal. While the buffer effect of boric acid and the bulk pH value indicate that no $Co(OH)_2$[24] precipitates will be formed. The intermediate was mainly composed of $CoOH^+_{(ads)}$ (Eq. 3)[37] and the intermediate will adsorbed on the electrode surface.

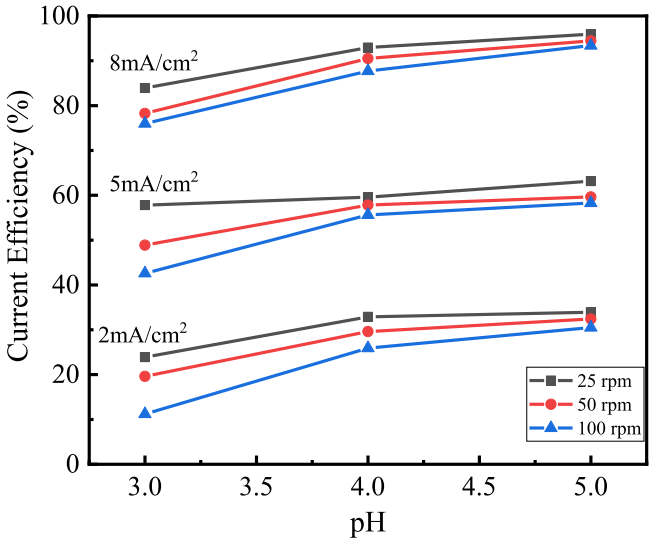


Figure 5. The current efficiency of the deposit as a function of pH, rotation rate, and applied current density, as calculated from potentiostatic deposition onto a Co-coated Pt RDE. The deposition time is 600 s and Co foil is used as anode.

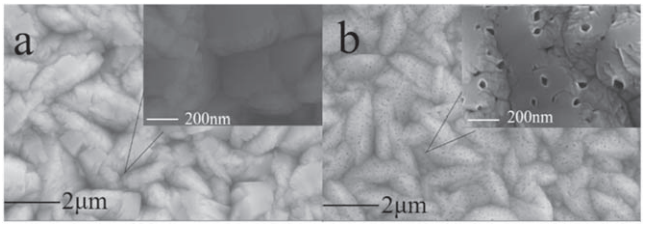


Figure 6. SEM pictures of Co deposition coating at −1.30 V for 5 min in the bath without (a) and with (b) 10ppm MG. The deposition bath consists of 0.4 mol l^{-1} $CoSO_4$, 0.5 mol l^{-1} H_3BO_3 and 10 mM Cl^-. Co-coated Cu used as working electrode and Co foil used as counter electrode at 100 rpm.

$$Co^{2+} + OH^- \rightarrow CoOH^+_{ads} \quad [3]$$

The influence of pH on current efficiency.—The effect of MG as an acceleration for HER would still result in an increasing of the local pH, especially at the bottom of the vias. Therefore, the current efficiency (CE) for Co^{2+} reduciton with different pH was studied, as shown in Fig. 5. The deposition current density was selected as 2 mA·cm^{-2}, 5 mA·cm^{-2}, and 8 mA·cm^{-2} respectively and the corresponding potential transients were shown in Fig. S7. The CE for the deposit was calculated by anodic stirring charge. It can be seen the proportion of current for Co^{2+} reduction increased gradually with the increasing of total current density or decreasing of rotation, resulting in higher CE. Although the concentration of Co^{2+} is significantly greater than that of H^+ in the deposition bath, the CE of H^+ reduction is primary at low current density. From Fig. 5, when current density was 2 mA cm^{-2}, the CE of Co^{2+} reduction was as low as 10% at pH 3. Under these conditions, changing either the reduction rate or transport rate of H^+ has noticeable effects on the CE of Co^{2+} deposition.[29,39,40] In addition to the concentration effect, proton reduction kinetics was confirmed to be faster than that of cobalt deposition at lower overpotential.[49] Therefore, the effect of MG on the proton reduction process should also be taken into account at lower overpotential deposition, especially at lower pH.

The surface morphology of cobalt deposit.—The surface morphology was analyzed to verify the acceleration of MG on HER, as shown in Figs. 6 and S5. SEM images were obtained with and without the existence of the additive MG. As can be seen from Fig. 6a, the Co grain are densely stacked blocks. When the MG concentration is 10 ppm, the grain size becomes smaller and there are much more nanoscale voids on the grain, as shown in the Fig. 6b. It can be concluded that the addition of MG results in the grain refinement because of the inhibition of MG on cobalt deposition. Meanwhile, the MG act as the accelerator for hydrogen evolution which make hydrogen bubble increase and overflow from the coating surface. However, the voids in the plating layer have a negative effect on the physical and chemical properties of the Co deposition and proper wetting agent should be introduced. Therefore, 20 ppm MPS and 10 mM chloride ion were added in the bath in order to obtain cobalt coating with high quality, as shown in Figs. S6a and S6b. After optimizing plating composition and process parameters, a compact uniform film without nanoscale voids can be obtained with or without stirring. The further verification of the contact angle under water was shown in Figs. S6c and S6d and the plating layer was placed face down under water and with a bubble attaching to the surface. The contact angle was 139° and 147° respectively with or without stirring, which means the released hydrogen during the entire reduction process can be quickly removed from the surface.

Electroplating.—The electroplating test was carried out in order to observe the influence of MG and pH on the bottom-up growth of cobalt. The test condition in filling experiments was the same as that in electroanalytical experiments except for stirring way. The ohmic losses difference caused by different stirring way can be neglected (Fig. S8). Electroplating test were investigated at the potential range of −1.19 V and −1.22 V since superconformal electrodeposition could be achieved due to intense pH and strong inhibition of MG. As shown in Fig. 7, there is almost no cobalt reduced when the cathode potential is −1.19 V. Referring to the previous voltammetry curve, when the external potential at −1.19 V, the main electrochemical reaction in the via is hydrogen ion reduction. When the deposition potential is −1.20 V, it can be seen that the deposition layer begins to appear at the bottom and side wall. According to Moffat's description of the metal filling process in TSV,[28,50,51] the blocking effect caused by the coverage of inhibitors on the electrode makes the deposition efficiency of metal on the sidewall higher than that on

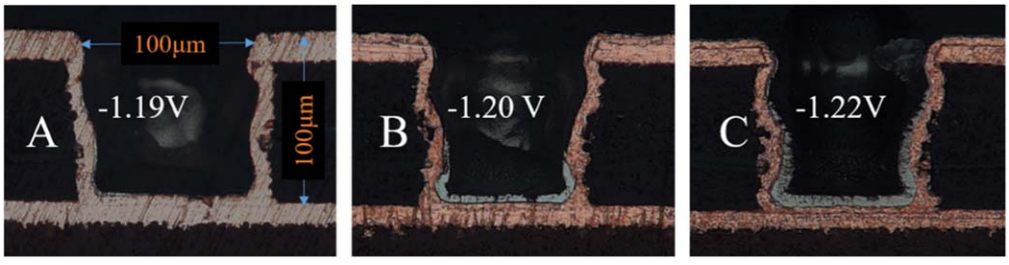


Figure 7. Cross-sectional optical microscope pictures at different potential for 2 h, (a) −1.19 V, (b) −1.20 V, (c) −1.22 V. The ratio of thickness to radial dimension is 100 μm×100 μm. The plating bath consists of 0.4 mol l^{-1} $CoSO_4$, 0.5 mol l^{-1} H_3BO_3,10 mM Cl^-, 20 ppm MPS and 10 ppm MG at pH 4.

the surface, which can form a passive-active transition. From the Fig. 7(b), it can be seen that the rapid deposition rates take place in the lower portion of the via, while the sidewall and outside surface remain in the passive state. What's more, the boundary of the passive-active transition gradually rises at more positive potential in Fig. 7(c). The transition forms mainly due to the difference of the vias coverage caused by the breakdown and diffusion-limited process of the MG.[28,52]

From the Fig. 2, the threshold value of the potential for cobalt deposition caused by suppression breakdown, is reflected in the dependence on MG concentration which related to the flux of MG to the surface. The distribution of the transition boundary is also controlled by the concentration of the bulk additive, as shown in the Fig. S9. the higher MG concentration was added, the lower active branch was exhibited in the bottom of the feature.

Although boric acid is added to the bath as buffer, it can be seen from the Fig. 5 that the deposition efficiency of the same pH value at different rotational speeds is significantly different, especially at a lower pH value. Next, the filling conditions of cobalt metal under different pH values were studied, as shown in Fig. 8. When the bath was adjusted the pH to 3 by dilute sulfuric acid, almost nothing is deposited in the feature, as shown in Fig. 8(a). This indicates that the reaction occurring at this condition is mainly the reduction of H^+, and the absorption of MG makes no cobalt deposit in the feature. After the pH value of the bath was up to 3.5, thin film was deposited on the central bottom rather than the bottom corners of the feature, as shown in Fig. 8(b). This model of deposition appears to differ from the higher deposition rate on the sidewall, which is based on the passive-active transition model. In other words, the MG inhibition effect still exist in the feature, and the cobalt deposit at the central bottom may be caused by some accelerant. In the absence of MG images (Fig. S10), the thickness of Co deposit in the feature, such as the bottom and the surrounding field, is nearly the same as that on the surface. Therefore, the local accelerant was probably regard as the formation of $CoOH^+_{ads}$ at the bottom,[53] which caused by the rising in pH, rather than the breakdown of the MG. A local pH increases owing to hydrogen ion consumption and mass transport limitations,[37] especially when MG is adsorbed at the bottom.

With the bulk pH increasing in Figs. 8(d), 8(e), the thickness of the bottom deposit increases gradually. No cobalt deposition coating means the existence of MG on the outside surface. The increase of pH value to 4.5 has different effects on the passive and active transition, compared with the model of pH 4 in the bath. The thickness of deposit on the upper sidewall increased might be due to the shrinking coverage of MG, which is caused by the substitution of $CoOH^+_{ads}$ on the recessed surface of the passive zone. While the lower proton concentration at pH 4.5 leads to higher deposit current efficiency and thermodynamically enhanced deposition especially when MG was broken down from the bottom corner of the active zone.

In order to achieve the superfilling in the vias, the PCB was deposited at a potential of −1.20 V for different times. After 1 h in Fig. 9(a), Co coating started to deposit from the bottom corner when deposition transition from passive to active because of the local breakdown of MG controlled by diffusion-limited process[29]al. As the deposition time extended to 2 h, deposition coating gradually covered the entire bottom. Localized breakdown on the sidewalls is a transient state that subsequently converts to bottom-up filling like. The formation of local pH gradient is a process of time accumulation because the mass transport of proton is faster than that of suppressor. The coupling of MG depletion effects in feature filling with the local increasing pH could provide the explanation for the subsequent shift from the passive to active zone at the bottom. When different rotation speeds and concentration are applied at −1.20 V, the filling of metal Co was completely different, as shown in Fig. S11. The phenomenon of conformal deposition at 0 rpm indicates the conformal growth of cobalt since effective MG concentration gradient was not established. As the convection intensifies, there is no cobalt deposit on the outer surface and sidewall of the via since effective MG concentration gradient based on diffusion-limited within the feature was established.[54]The passive-active profile should be steady-state in nature, which is reflected in the potential-dependent incorporation balanced by the suppressor depletion gradient in the deposit.[52] However, this transition state gradually expands over time [Fig. 9(c)], which may be due to the disruption of inhibition caused by pH evolution, which is related to time evolution. After the stable passive-active transition zone is completed, the bottom-up growth process proceeds smoothly as shown in Figs. 9(d) and 9(e).

According to the deposition process provided above, the superfilling growth of cobalt in PCB board is realized in the Fig. 10(a). The schematic diagram of the filling process was exhibited in the Fig. 10(b) and induction-diffusion-adsorption filling mechanism was proposed, which consists of the induction effect of MG on HER as well as the diffusion-adsorption effect[55] caused by the MG concentration gradient and pH gradient within the blind vias.

At the initial state, the cathode surface covered with inhibitor MG which suppress the Co reduction and promote the HER. During the electrodeposition, the deposit suppression gradient was formed because of the diffusion-limited of MG. The passive-active transition zone come into being, and then the zone grows until it covers the entire trench due to the induction HER of MG. Meantime, the effective pH gradient is also established due to the mass transport limitation of hydrogen ion, resulting that the current efficiency at the bottom is higher, as shown in stage b in Fig. 10. Compared with the formation of inhibition gradient, the disruption of MG caused by HER is weaker, which is more conducive to the bottom-up superfilling process. In summary, the induction-diffusion-adsorption mechanism, namely diffusion-adsorption effect caused by MG gradient and pH gradient as well as induction effect of MG on HER, was proposed to illustrate the bottom-up growth of cobalt.

Conclusions

The additive Malachite Green (MG) was used as an effective suppressor for cobalt deposition. The suppressing effect of MG on cobalt electrodeposition was investigated according to cyclic voltammetry and the mass transport of MG is diffusion-controlled. The induction acceleration of MG on HER was obtained by coupling EQCM and cyclic voltammetry as well as surface morphology was used to confirm the acceleration of MG on HER, while the influence of pH on current efficiency was also studied. The formation of hydroxide ions at the electrode surface caused by the HER which leaded to the formation of $CoOH^+_{(ads)}$. The induction-diffusion-adsorption mechanism was proposed to illustrate the bottom-up growth of cobalt. The MG gradient formed within blind vias due to the limitation of mass transport resulting the inhibition difference at

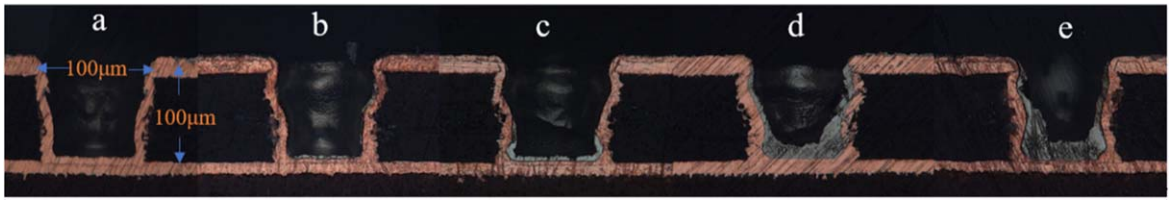


Figure 8. Cross-sectional optical microscope pictures of different pH of Co filling at −1.20 V, (a) 3, (b) 3.5, (c) 4 (d) 4.5 (e) 5. The plating bath consists of 0.4 mol l^{-1} $CoSO_4$, 0.5 mol l^{-1} H_3BO_3, 10 mM of Cl^-, 20 ppm MPS and 10 ppm MG.

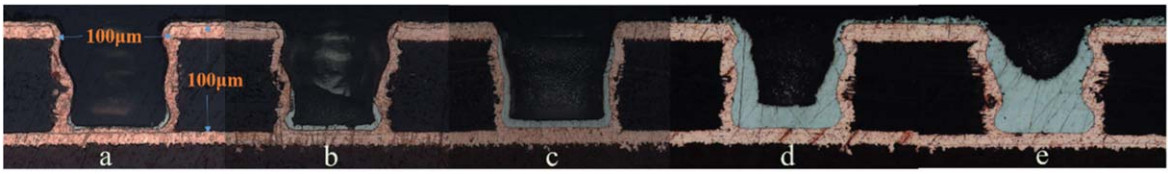


Figure 9. Cross-sectional optical microscope pictures of time evolution of Co filling at −1.20 V, (a) 1 h, (b) 2 h, (c) 4 h (d) 8 h (e)16 h. The plating bath consists of 0.4 mol l^{-1} $CoSO_4$, 0.5 mol l^{-1} H_3BO_3, 10 mM of Cl^-, 20 ppm MPS and 10 ppm MG at pH 4.

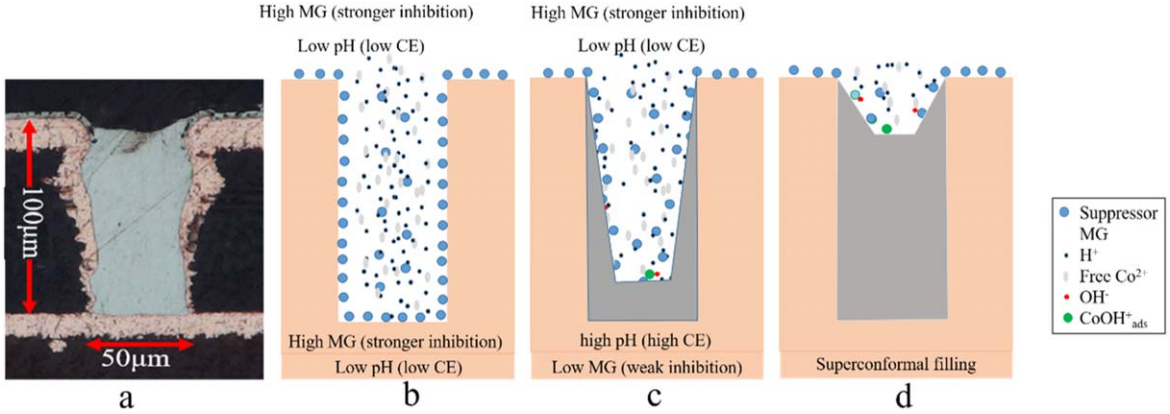


Figure 10. Cross-sectional optical microscope pictures of Co superfilling from the solution consisted of 0.4 mol l^{-1} $CoSO_4$, 0.5 mol l^{-1} H_3BO_3, 10 mM of Cl^-, 20 ppm MPS and 15 ppm MG. (a) and the simulation schematic diagram of the progressively filling process (b)–(d).

different parts (the growth rate at the bottom is higher than that of the opening), while the pH gradient and the formation of $CoOH^+_{(ads)}$ caused by MG induction is beneficial to the Cobalt filling in the feature.

Acknowledgments

This work was supported by the National Nature Science Foundation of China (No. 21972037) and State Key Laboratory of Urban Water Resources and Environment (Harbin Institute of Technology) (No.2021TS07).

ORCID

Maozhong An https://orcid.org/0000-0003-1215-0519

References

1. P. C. Andricacos, C. Uzoh, J. O. Dukovic, J. Horkans, and H. Deligianni, "Damascene copper electroplating for chip interconnections." *IBM J. Res. Dev.*, **42**, 567 (1998).
2. C. Adelmann et al., "IEEE international interconnect technology conference/ advanced metallization conference (IITC/AMC)." *IEEE International Interconnect Technology Conference IITC, San J*(ose, CA)(Ieee)p. 173 (2014).
3. S. Brandstetter, V. Carreau, S. Maîtrejean, M. Verdier, and M. Legros, "Grain morphology of Cu damascene lines." *Microelectron. Eng.*, **87**, 383 (2010).
4. P.-I. Wang, M. D. Frey, M. Washington, S. Nayak, and T.-M. Lu, "Resistivity of sub-50 nm copper lines epitaxially grown on Si(100) substrate." *Thin Solid Films*, **520**, 6106 (2012).
5. W. Zhang, S. H. Brongersma, Z. Li, D. Li, O. Richard, and K. Maex, "Analysis of the size effect in electroplated fine copper wires and a realistic assessment to model copper resistivity." *J. Appl. Phys.*, **101**, 11 (2007).
6. D. Josell, S. H. Brongersma, and Z. Tőkei, "Size-dependent resistivity in nanoscale interconnects." *Annu. Rev. Mater. Res.*, **39**, 231 (2009).
7. D. Gall, "Electron mean free path in elemental metals." *J. Appl. Phys.*, **119**, 5 (2016).
8. J. Kelly et al., "Annealing and impurity effects in Co thin films for MOL contact and BEOL metallization." *J. Electrochem. Soc.*, **166**, D3100 (2018).
9. V. Kamineni et al., "IEEE international interconnect technology conference/ advanced metallization conference (IITC/AMC)." *IEEE International Interconnect Technology Conference IITC*(San Jose, CA)(Ieee)p. 105 (2016).
10. R. L. Graham, G. B. Alers, T. Mountsier, N. Shamma, S. Dhuey, S. Cabrini, R. H. Geiss, D. T. Read, and S. Peddeti, "Resistivity dominated by surface scattering in sub-50 nm Cu wires." *Appl. Phys. Lett.*, **96**, 3 (2010).
11. N. A. Lanzillo, P. Bhosale, C. Lavoie, D. J. Dechene, R. R. Robison, and K. Choi, "Spin-dependent electron scattering in cobalt interconnects." *J. Phys. D: Appl. Phys.*, **52**, 8 (2019).
12. M. H. van der Veen et al., "in 24th Workshop on Materials for Advanced Metallization held Jointly with the International Interconnect Technology (IITC) Conference." *IEEE International Interconnect Technology Conference IITC* (Grenoble, FRANCE)(Ieee)p. 25 (2015).
13. J. Kelly et al., "IEEE international interconnect technology conference/advanced metallization conference (IITC/AMC)." *IEEE International Interconnect Technology Conference IITC*(San Jose, CA)(Ieee)p. 40 (2016).
14. M. Darques, A. Encinas, L. Vila, and L. Piraux, "Controlled changes in the microstructure and magnetic anisotropy in arrays of electrodeposited Co nanowires induced by the solution pH." *J. Phys. D: Appl. Phys.*, **37**, 1411 (2004).
15. J.-Y. Lin, J.-H. Liao, and S.-W. Chou, "Cathodic electrodeposition of highly porous cobalt sulfide counter electrodes for dye-sensitized solar cells." *Electrochim. Acta*, **56**, 8818 (2011).
16. M. P. M. Kaninski, V. M. Nikolic, G. S. Tasic, and Z. L. Rakocevic, "Electrocatalytic activation of Ni electrode for hydrogen production by electro-deposition of Co and V species." *Int. J. Hydrog. Energy*, **34**, 703 (2009).
17. R. P. Li, Y. Q. Li, P. X. Yang, P. H. Ren, D. Wang, X. Y. Lu, R. Y. Xu, Y. H. Li, J. M. Xue, J. Q. Zhang, M. Z. An, J. Y. Ma, B. Wang, H. K. Liu, and S. X. Dou, "Synergistic interfaceengineering and structural optimization of non-noble metal telluride-nitrideelectrocatalysts for sustainably overall seawater electrolysis." *Appl. Catal. B-Environ.*, **318**, 12 (2022).
18. B. Pan, Y. Yang, Y. Zhang, and Q. Zhang, "Effects of saccharin and tetramethy-lammonium bromide on the microstructure and microhardness of thick cobalt electrodeposits." *Surf. Coat. Technol.*, **329**, 49 (2017).
19. A. C. Frank and P. T. A. Sumodjo, "Electrodeposition of cobalt from citrate containing baths." *Electrochim. Acta*, **132**, 75 (2014).
20. M. S. Bhuiyan, B. J. Taylor, M. Paranthaman, J. R. Thompson, and J. W. Sinclair, "Microstructure and magnetic properties of electrodeposited cobalt films." *J. Mater. Sci.*, **43**, 1644 (2008).
21. L. Wang, Y. Lin, Z. Zeng, W. Liu, Q. Xue, L. Hu, and J. Zhang, "Electrochemical corrosion behavior of nanocrystalline Co coatings explained by higher grain boundary density." *Electrochim. Acta*, **52**, 4342 (2007).
22. D. Hamulić, I. Milošev, and D. Lützenkirchen-Hecht, "The effect of the deposition conditions on the structure, composition and morphology of electrodeposited cobalt materials." *Thin Solid Films*, **667**, 11 (2018).
23. Y. Tsuru, M. Nomura, and F. R. Foulkes, "Effects of boric acid on hydrogen evolution and internal stress in films deposited from a nickel sulfamate bath." *J. Appl. Electrochem.*, **32**, 629 (2002).
24. M. A. Rigsby, T. A. Spurlin, and J. D. Reid, "The Multi-Functional Role of Boric Acid in Cobalt Electrodeposition and Superfill." *J. Electrochem. Soc.*, **167**, 8 (2020).
25. J. P. Hoare, "On the role of boric acid in the Watts bath." *J. Electrochem. Soc.*, **133**, 2491 (1986).

26. C. H. Lee, J. E. Bonevich, J. E. Davies, and T. P. Moffat, "Superconformal Electrodeposition of Co and Co-Fe Alloys Using 2-Mercapto-5-benzimidazolesulfonic Acid." *J. Electrochem. Soc.*, **156**, D301 (2009).
27. M. A. Rigsby, L. J. Brogan, N. V. Doubina, Y. Liu, E. C. Opocensky, T. A. Spurlin, J. Zhou, J. D. Reid, and S. Electrochem, *in 232nd Fall Meeting of the-Electrochemical-Society (ECS), ECS Transactions, National Harbor, MD*p. 767 (2017).
28. D. Josell, M. Silva, and T. P. Moffat, "Superconformal bottom-up cobalt deposition in high aspect ratio through silicon vias." *J. Electrochem. Soc.*, **163**, D809 (2016).
29. A. Joi, R. Akolkar, and U. Landau, "Additives for bottom-up copper plating from an alkaline complexed electrolyte." *J. Electrochem. Soc.*, **160**, D3001 (2013).
30. C. Madore, M. Matlosz, and D. Landolt, "Blocking inhibitors in cathodic leveling.1. Theoretical analysis." *J. Electrochem. Soc.*, **143**, 3927 (1996).
31. K. G. Jordan and C. W. Tobias, "The effect of inhibitor transport on leveling in electrodeposition." *J. Electrochem. Soc.*, **138**, 1251 (1991).
32. J. J. Kelly, C. Y. Tian, and A. C. West, "Leveling and microstructural effects of additives for copper electrodeposition." *J. Electrochem. Soc.*, **146**, 2540 (1999).
33. J. Wu, F. Wafula, S. Branagan, H. Suzuki, and J. van Eisden, "Mechanism of cobalt bottom-up filling for advanced node interconnect metallization." *J. Electrochem. Soc.*, **166**, D3136 (2018).
34. Q. Huang, T. W. Lyons, and W. D. Sides, "Electrodeposition of Cobalt for Interconnect Application: Effect of Dimethylglyoxime." *J. Electrochem. Soc.*, **163**, D715 (2016).
35. Y. J. Han, X. Zhang, and G. W. Leach, "Shape control of electrodeposited copper films and nanostructures through additive effects." *Langmuir*, **30**, 3589 (2014).
36. X. C. Ma, Y. Q. Li, P. X. Yang, J. Q. Zhang, and M. Z. An, "Influence of suppressing additive malachite green on superconformal cobalt electrodeposition." *J. Electroanal. Chem.*, **921**, 10 (2022).
37. M. A. Rigsby, L. J. Brogan, N. V. Doubina, Y. Liu, E. C. Opocensky, T. A. Spurlin, J. Zhou, and J. D. Reid, "The Critical role of pH gradient formation in driving superconformal cobalt deposition." *J. Electrochem. Soc.*, **166**, D3167 (2018).
38. K. Y. Lin and H. A. Chang, "Ultra-high adsorption capacity of zeolitic imidazole framework-67 (ZIF-67) for removal of malachite green from water." *Chemosphere*, **139**, 624 (2015).
39. D. R. Gabe, "The role of hydrogen in metal electrodeposition processes." *J. Appl. Electrochem.*, **27**, 908 (1997).
40. I. Flis-kabulska, "Electrodeposition of cobalt on gold during voltammetric cycling." *J. Appl. Electrochem.*, **36**, 131 (2005).
41. T. P. Moffat and D. Josell, "Extreme bottom-up superfilling of through-silicon-vias by damascene processing: suppressor disruption, positive feedback and turing patterns." *J. Electrochem. Soc.*, **159**, D208 (2012).
42. S. K. Kim, J. E. Bonevich, D. Josell, and T. P. Moffat, "Electrodeposition of ni in submicrometer trenches." *J. Electrochem. Soc.*, **154**, D443 (2007).
43. V. Grozovski, S. Vesztergom, G. G. Láng, and P. Broekmann, "Electrochemical hydrogen evolution: H+or H2O reduction ? a rotating disk electrode study." *J. Electrochem. Soc.*, **164**, E3171 (2017).
44. D. Gabe, "The role of hydrogen in metal electrodeposition processes." *J. Appl. Electrochem.*, **27**, 908 (1997).
45. X. Ma, Y. Li, P. Yang, J. Zhang, and M. An, "Influence of suppressing additive malachite green on superconformal cobalt electrodeposition." *J. Electroanal. Chem.*, **921** (2022).
46. Y. Umasankar, A. P. Periasamy, and S. M. Chen, "Electrocatalysis and simultaneous determination of catechol and quinol by poly(malachite green) coated multiwalled carbon nanotube film." *Anal. Biochem.*, **411**, 71 (2011).
47. P. A. Jacques, V. Artero, J. Pecaut, and M. Fontecave, "Cobalt and nickel diimine-dioxime complexes as molecular electrocatalysts for hydrogen evolution with low overvoltages." *Proc. Natl. Acad. Sci. U. S. A.*, **106**, 20627 (2009).
48. C. C. McCrory, C. Uyeda, and J. C. Peters, "Electrocatalytic hydrogen evolution in acidic water with molecular cobalt tetraazamacrocycles." *J. Am. Chem. Soc.*, **134**, 3164 (2012).
49. V. Grozovski, P. Moreno-García, E. Karst, M. de Jesús Gálvez-Vázquez, A. Fluegel, S. Kitayaporn, S. Vesztergom, and P. Broekmann, "Operando laser scattering: probing the evolution of local pH changes on complex electrode architectures." *J. Electrochem. Soc.*, **168**, 7 (2021).
50. D. Josell, C. Witt, T. J. E. Moffat, and S.-S. Letters, "Osmium barriers for direct copper electrodeposition in damascene processing." **9**, C41 (2005).
51. D. Josell and T. P. Moffat, "Superconformal copper deposition in through silicon vias by suppression-breakdown." *J. Electrochem. Soc.*, **165**, D23 (2018).
52. T. M. Braun, S. H. Kim, H. J. Lee, T. P. Moffat, and D. Josell, "Superconformal nickel deposition in through silicon vias: experiment and prediction." *J. Electrochem. Soc.*, **165**, D291 (2018).
53. V. P. Graciano, U. Bertocci, and G. R. Stafford, "In-situ stress measurements during cobalt electrodeposition." *J. Electrochem. Soc.*, **166**, D3246 (2019).
54. S. J. Banik, K. K. Rao, and R. Akolkar, "Determination of the diffusion-adsorption properties of polymeric electrolyte additives using an additive injection method implemented on a rotating disc electrode." *J. Electrochem. Soc.*, **163**, E241 (2016).
55. R. Akolkar and V. Dubin, "Pattern density effect on the bottom-up fill during damascene copper electrodeposition - Analysis of the suppressor concentration field effect." *Electrochem. Solid State Lett.*, **10**, D55 (2007).